

Causal Inference in Reinforcement Learning

B.Tech Project – Final Presentation

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Reinforcement Learning: Basic Interaction Framework

Reinforcement Learning formalizes how an agent interacts with an environment to make decisions based on feedback signals.

Core components:

- State s : information available to the agent
- Action a : choice sampled from policy $\pi_{\theta}(a|s)$
- Reward r : feedback signal used for learning

Single-step setting:

- Objective reduces to maximizing:

$$J(\pi) = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi} [r(s, a)]$$

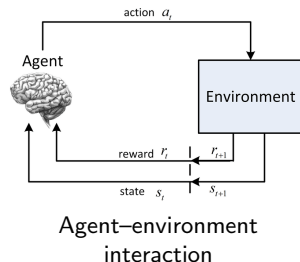


Figure reference: Adapted from <https://doi.org/10.48550/arXiv.1712.00162>

Policy Gradient: Core Idea

Parametric policy: $\pi_{\theta}(a|s)$

Goal: maximize expected return

$$J(\pi_{\theta}) = \mathbb{E}_{s \sim \mathcal{D}, a \sim \pi_{\theta}}[r(s, a)]$$

Policy Gradient Theorem:

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{\pi_{\theta}} \left[\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi}(s, a) \right]$$

Key intuition:

- Move parameters in direction that increases probability of high-reward actions.
- Works for discrete (like bandits, queuing) and continuous (like robotics) action spaces both.

Regret and Sample Complexity

Regret:

$$\text{Regret}(T) = \sum_{t=1}^T (V(a^*) - V(a_t))$$

Quantifies cumulative loss from taking suboptimal actions.

Sample Complexity (SC): Number of episodes needed for an algorithm to achieve near-optimal performance (e.g., ϵ -optimal with high probability).

Regret Bound for General Multi-Armed Bandits

Theorem

For a general stochastic MAB system satisfying smoothness and Polyak–Łojasiewicz (PL) conditions, the cumulative regret after T rounds satisfies:

$$R_T \leq \frac{A}{\mu_{\text{PL}} \eta^*} + \frac{L \eta^* T \sigma^2}{2\mu_{\text{PL}}} = \frac{\sqrt{2}}{\mu_{\text{PL}}} \sqrt{AL \sigma^2 T},$$

where σ^2 is the variance of the reward process.

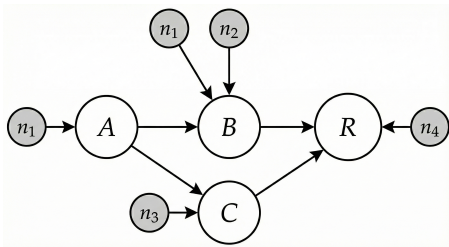
Proof Sketch.

- PL condition to convert optimization error into a quadratic form.
- High-order statistical concentration bounds on noisy gradients.
- Optimize the learning rate η^* to balance bias–variance terms.

Implication: Regret $\Rightarrow$ Sample Complexity

- Lower regret bounds directly imply fewer suboptimal pulls.
- Since $R_T = \sum_t (\mu^* - \mu_{a_t})$, smaller regret means the agent identifies the optimal arm faster.
- Thus, **lower variance $\Rightarrow$ lower regret $\Rightarrow$ lower sample**

Causal Representations



- Structural Causal Model (SCM)
- Nodes, Directed Edges, Structural Equations
- A influences both B and C
- A influences R too
- Minimal structure required to derive counterfactuals.

Variance Reduction with Counterfactual Baselines

Proposition (Counterfactual baselines reduce gradient variance).

Let the policy-gradient estimator be

$$g = (R(A) - b) \nabla_{\theta} \log \pi_{\theta}(A|X).$$

If $b_{CF}(X, U)$ is a counterfactual baseline computed under the same exogenous noise U , then:

$$\text{Var}[g_{CF}] \leq \text{Var}[g_{\text{std}}],$$

with equality only when b_{CF} adds no information.

Intuition:

- $b_{CF}(X, U)$ cancels exogenous noise shared across counterfactual worlds.
- Removes variance due to uncontrollable factors, leaving only true action-based variation.

Implications

- Lower gradient variance $\Rightarrow$ **tighter regret bounds.**
- Lower regret $\Rightarrow$ **lower sample complexity.**
- Hence the agent reaches near-optimal actions in **fewer steps.**

Causal Policy Gradients

Counterfactual Baseline:

$$b_{CF}(X, U) = f(X, U)$$

Causal Advantage:

$$A_c = R(A) - b_{CF}(X, U)$$

Causal REINFORCE

Gradient update uses the causal advantage:

$$\nabla_{\theta} J \approx A_c \nabla_{\theta} \log \pi_{\theta}(A | X)$$

Remove variance due to exogenous noise, making the gradient more stable without introducing bias.

C-PPO

Hybrid advantage mixes causal and classical PPO baselines:

$$\begin{aligned} A_c^{\text{PPO}} &= \alpha (R(A) - b_{CF}(X, U)) \\ &+ (1 - \alpha) (R(A) - b_{PPO}(A)) \end{aligned}$$

Causal REINFORCE — High-Level Algorithm

Algorithm 1 Causal Policy Gradient (High-Level)

Initialize policy parameters θ

for each episode **do**

Collect trajectory (s_t, a_t, r_t)

for each step t **do**

Abduction: infer U_R from r_t

CF Simulation: compute $r_t(a')$ for $a' \neq a_t$

Compute CF baseline b_{CF}

$A_{CF} \leftarrow r_t - b_{CF}$

$\theta \leftarrow \theta + \alpha A_{CF} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t)$

end for

end for

Causal Baseline Variants

Given counterfactual rewards $\{R(a') : a' \neq A\}$ and policy $\pi(a'|X)$:

① **Mean of all non-taken actions:**

$$b_1 = \frac{1}{K-1} \sum_{a' \neq A} R(a')$$

② **Policy-weighted mean:**

$$b_2 = \sum_{a' \neq A} \pi(a'|X) R(a')$$

③ **Top- k mean (unweighted):**

$$b_3 = \frac{1}{k} \sum_{a' \in \text{Top-}k(R)} R(a')$$

Causal Baseline Variants

Continuing with top- k based baselines:

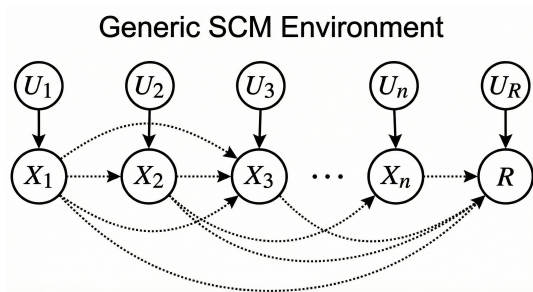
④ **Top- k policy-weighted mean:**

$$b_4 = \frac{\sum_{a' \in \text{Top-}k(R)} \pi(a'|X) R(a')}{\sum_{a' \in \text{Top-}k(R)} \pi(a'|X)}$$

⑤ **Top- k mean of $(\pi \cdot R)$:**

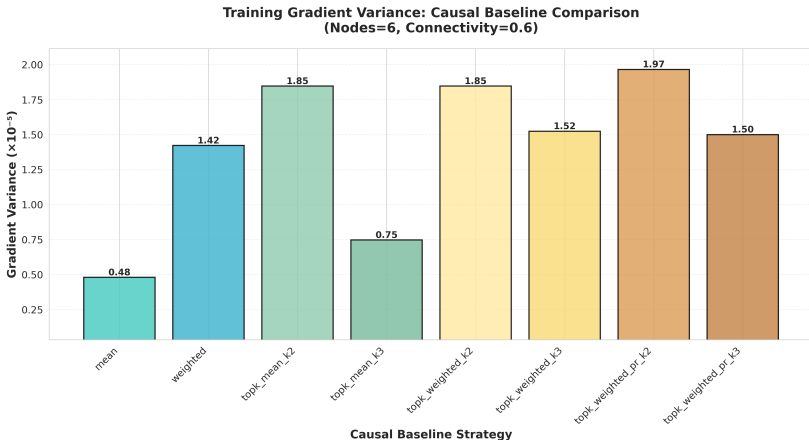
$$b_5 = \frac{1}{k} \sum_{a' \in \text{Top-}k(\pi(a'|X) R(a'))} \pi(a'|X) R(a')$$

Generic SCM Environment



- Random acyclic causal graph (edge probability 0.3/0.6)
- Each node uses a randomly assigned structural equation (linear / poly / sigmoid / mix)
- Exogenous noise sampled from Gaussian / Uniform / Laplace
- Actions as predefined do- interventions

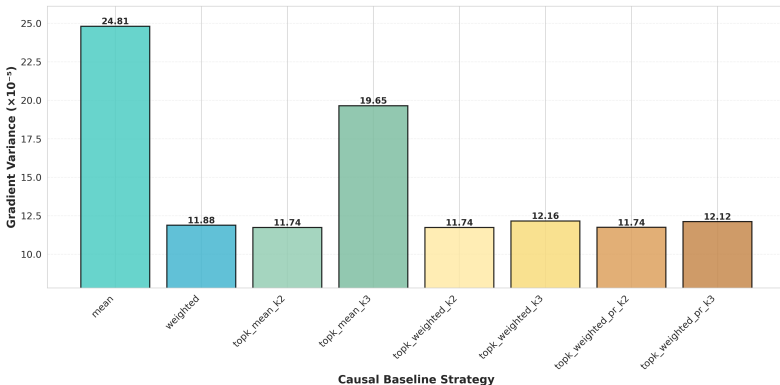
Exp1 - Gradient Variance Comparison



Lower is better (more stable). Vanilla REINFORCE: 1.99×10^{-5} | Best causal: 0.48×10^{-5} (mean) | Reduction: 75.9%

Exp1 - Gradient Variance Comparison

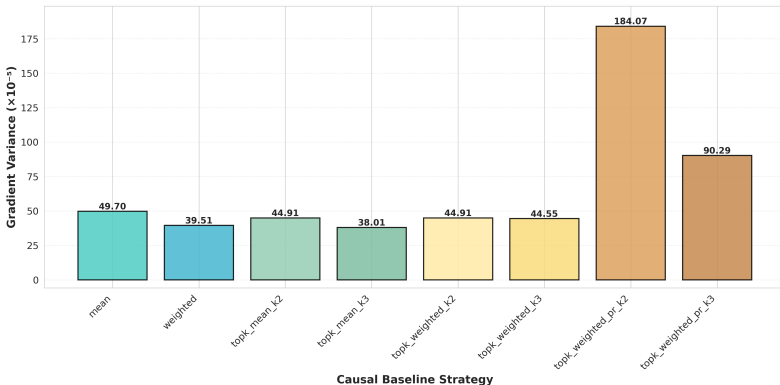
Training Gradient Variance: Causal Baseline Comparison
(Nodes=7, Connectivity=0.3)



Lower is better (more stable). Vanilla REINFORCE: 740.92×10^{-5} | Best causal: 11.74×10^{-5} (topk_mean_k2) | Reduction: 98.4%

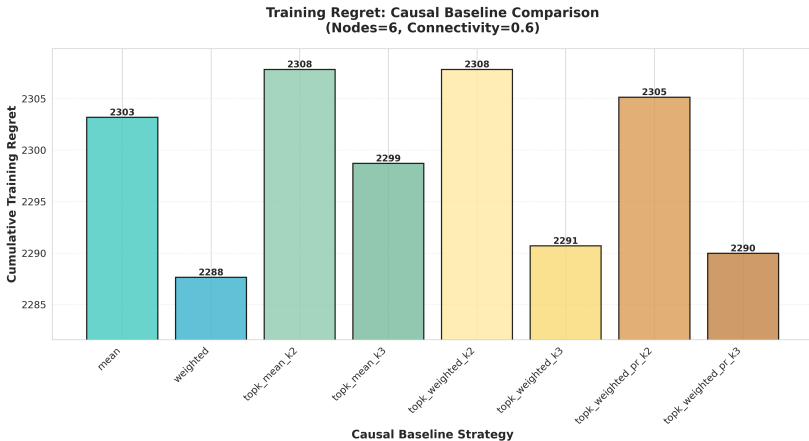
Exp1 - Gradient Variance Comparison

Training Gradient Variance: Causal Baseline Comparison
(Nodes=7, Connectivity=0.6)



Lower is better (more stable). Vanilla REINFORCE: 448.27×10^{-5} | Best causal: 38.01×10^{-5} (topk_mean_k3) | Reduction: 91.5%

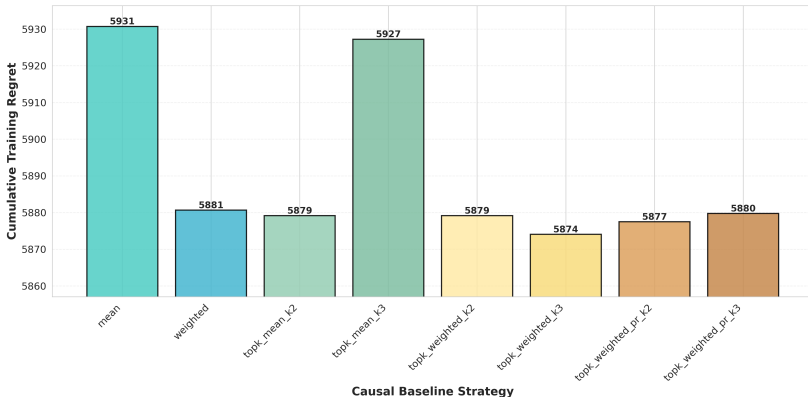
Exp1 - Regret Comparison



Lower is better. Vanilla REINFORCE: 2311 | Best causal: 2288 (weighted) | Improvement: 1.0%

Exp1 - Regret Comparison

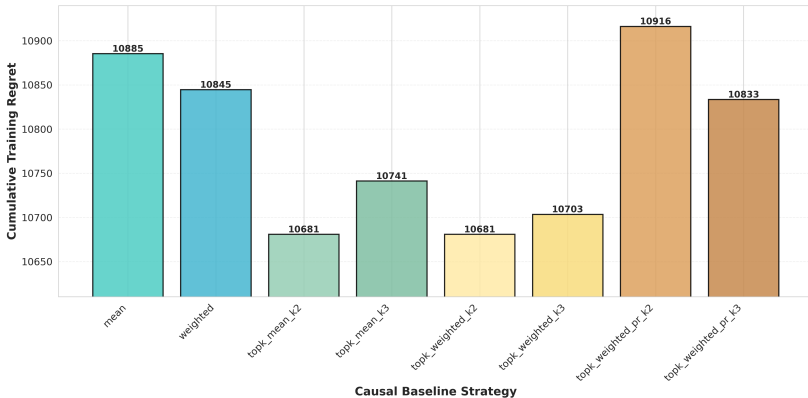
Training Regret: Causal Baseline Comparison
(Nodes=7, Connectivity=0.3)



Lower is better. Vanilla REINFORCE: 6308 | Best causal: 5874 (topk_weighted_k3) | Improvement: 6.9%

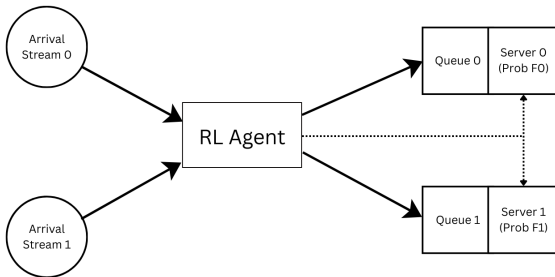
Exp1 - Regret Comparison

**Training Regret: Causal Baseline Comparison
(Nodes=7, Connectivity=0.6)**



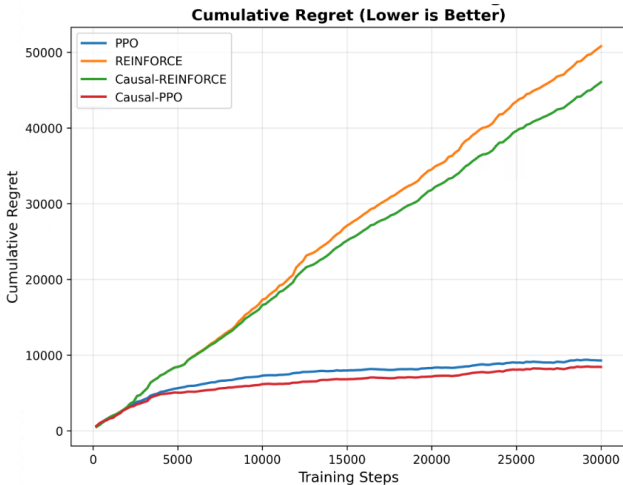
Lower is better. Vanilla REINFORCE: 11306 | Best causal: 10681 (topk_mean_k2) | Improvement: 5.5%

Parallel Queue SCM

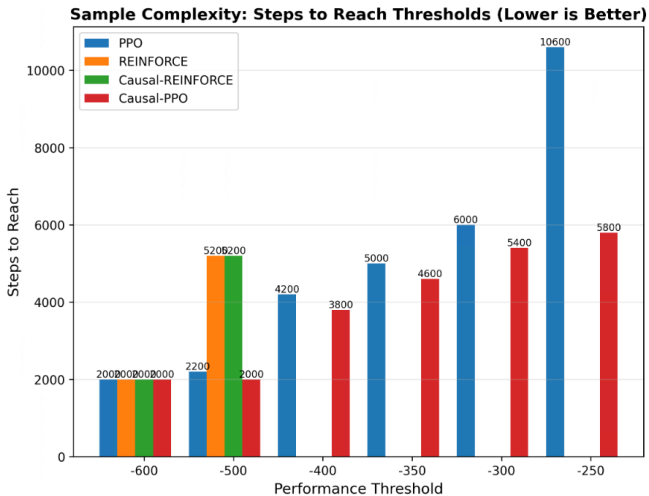


- Two parallel queues receiving stochastic arrivals and service outcomes.
- Agent decides how to route incoming jobs and which server to activate.
- Noise includes random arrivals and service successes/failures.

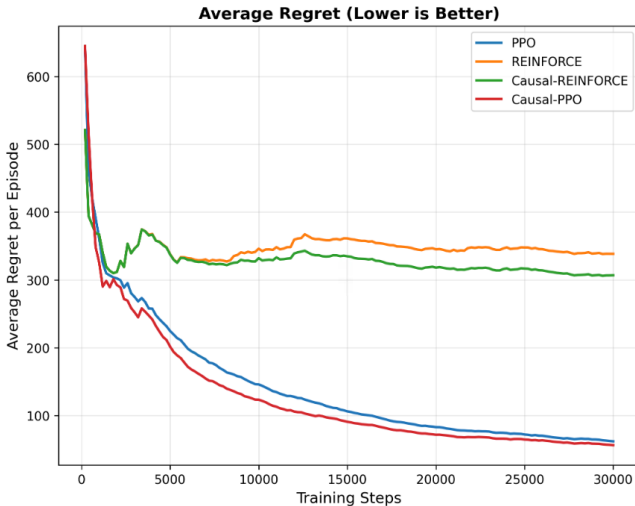
Exp2 - Cumulative Regret Comparison



Exp2 - Sample Complexity Comparison



Exp2 - Average Regret Comparison



Conclusion & Future Work

Summary

- Classical policy gradients suffer from variance and spurious correlations.
- Counterfactual baselines isolate true action impact and significantly reduce variance.
- Consistent improvements observed across both experimental setups.

Future Work

- Partial-observation and robustness settings
- Online SCM estimation
- Multi-agent / communication extensions

Target: ICML 2026
(Flagship A* conference in Machine Learning)

Abstract due Jan 23 (AoE), Full paper due Jan 28 (AoE)

Questions?

Thank you!

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